RACE
Regular Analysis Through Continuous Exercise

Differentiability : RACE-01

Single Correct Question

1. A function is defined as follows

$f(x)=\begin{cases} x^3 & ; \quad x^2<1 \\ x & ; \quad x^2\geq 1 \end{cases}$ is

(A) Continuous but not differentiable at $x = 1$

(B) Discontinuous but differentiable at $x = 1$

(C) Discontinuous but not differentiable at $x = 1$

(D) Continuous but differentiable at $x = 1$

MDY132

2. If $f(x)=\begin{cases} \ell nx & \text{for} \quad x\geq e^2 \\ ax+b & \text{for} \quad x<e^2 \end{cases}$ is differentiable at $x = e^2$, then find value of a and b.

(A) $a=\frac{3}{e^2}, b=1$ (B) $a=\frac{1}{e^2}, b=2$ (C) $a=\frac{1}{e^2}, b=1$ (D) $a=\frac{1}{e^2}, b=5$

MDY133

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3. If $f(x)=\begin{cases} x & , \quad x\leq 1 \\ x^2+bx+c & , \quad x>1 \end{cases}$ then find the values of b and c if $f(x)$ is differentiable at $x = 1$

(A) $b = -2, c = 1$ (B) $b = -2, c = -2$ (C) $b = -1, c = 2$ (D) $b = -1, c = 1$

MDY134

4. A function is defined as follows $f(x) = (x-1)|x-1|$.

(A) Continuous & not differentiable at $x = 1$

(B) Continuous & differentiable at $x = 1$

(C) Discontinuous & differentiable at $x = 1$

(D) Discontinuous & not differentiable at $x = 1$

MDY135

5. Which of the following functions is differentiable at $x = 0$?

(A) $\cos(|x|) + |x|$ (B) $\cos(|x|) - |x|$ (C) $\sin(|x|) + |x|$ (D) $\sin(|x|) - |x|$

MDY136

6. If $f(x) = x\left(\sqrt{x}-\sqrt{x+1}\right)$, then indicate the correct alternative(s):

(A) $f(x)$ is continuous but not differentiable at $x = 0$

(B) $f(x)$ is differentiable at $x = 0$

(C) $f(x)$ is not differentiable at $x = 0$

(D) none

MDY137

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Multiple Correct Question

7. If $f(x)=\begin{cases} ax^3+b, & \text{for } 0\le x\le 1 \\ 2\cos \pi x+\tan^{-1} x, & \text{for } 1<x\le 2 \end{cases}$ be the differentiable function at $x=1$, then find a and b. (where [.] denotes the greatest integer function)

(A) $a=\frac{1}{3}$ (B) $b=\frac{\pi}{4}-\frac{13}{6}$ (C) $a=\frac{1}{6}$ (D) $b=\frac{\pi}{4}-\frac{13}{3}$

MDY138

8. Select the correct statements-

(A) $f(x)=(x+1)^{1/3}$ is not differential at $x=-1$

(B) $f(x)=3|x-1|$ is not differential at $x=1$

(C) $f(x)=[x+1]$ is not differential at $x=0$ (where [.] denotes the greatest integer function)

(D) $f(x)=x^2+2x+1$ is not differential at $x=0$

MDY139

9. The function $f(x)=\begin{cases} x^3-1 & ; \quad 1<x<\infty \\ x-1 & ; \quad -\infty<x\le 1 \end{cases}$ is -

(A) continuous at $x=1$

(B) Differentiable at $x=1$

(C) Discontinuous at $x=1$

(D) Not differentiable at $x=1$

MDY140

10. A function is defined as follows

$$f(x)=\begin{cases} 1 & , \quad -\infty<x<0 \\ 1+|\sin x| & , \quad 0\le x<\frac{\pi}{2} \\ 2+\left(x-\frac{\pi}{2}\right)^2 & , \quad \frac{\pi}{2}\le x<\infty \end{cases}.$$

(A) Non-derivable at $x=0$ (B) continuous at $x=0$

(C) derivable and continuous at $x=\frac{\pi}{2}$ (D) derivable and discontinuous at $x=\frac{\pi}{2}$

MDY141

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	C	D	B	D	B	B,C	A,B,C	A,D	A,B,C

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RACE
Regular Analysis Through Continuous Exercise

Differentiability : RACE-02

Single Correct Question

1. A function $f:\mathbb{R} \to (0, \infty)$ satisfies the equation $f(x+y) = f(x).f(y)$. If $f(x) \neq 0$ for any x, is differentiable on $\mathbb{R}$ and $f'(0) = 2$ then find $f(x)$

(A) e^{-x} (B) e^{2x} (C) e^{x} (D) e^{-2x}

MDY142

2. A function $f:\mathbb{R}\to\mathbb{R}$ satisfies the condition $f\left(\frac{x+y}{k}\right)=\frac{f(x)+f(y)}{k}$ $(k \neq 0, 2)$. If $f(x)$ is differentiable on $\mathbb{R}$ and $f'(0) = m$ then find $f(x)$

(A) mx (B) $x^2 + mx$ (C) $mx + 1$ (D) $x^3 + mx$

MDY143

Multiple Correct Question

3. A function is defined as follows $f(x) = \begin{cases} [x] & ; \ 0 < x \leq 2 \\ 2x-2 & ; 2 < x < 3 \end{cases}$, where [.] denotes greatest integer function.

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(A) continuous and not differential at $x = 1$

(B) discontinuous and not differential at $x = 2$

(C) discontinuous and not differential at $x = 1$

(D) discontinuous and differential at $x = 2$

MDY144

4. The function $f(x) = \begin{cases} \frac{\pi}{4} + \tan^{-1} x, & |x| \leq 1 \\ \frac{1}{2}(|x|-1), & |x| > 1 \end{cases}$. Discuss the continuity and differentiability of $f(x)$:

(A) continuous on $R-\{1,-1\}$ (B) continuous on $R-\{1\}$

(C) Differentiable on $R-\{-1,1\}$. (D) Differentiable on $R-\{1\}$.

MDY145

5. Suppose f is a derivable function that satisfies the equation $f(x+y) = f(x) + f(y) + x^2y + xy^2$ for all real numbers x and y. Where $\lim_{x\to 0}\frac{f(x)}{x}=1$ then find

(A) $f(0) = 0$ (B) $f'(0) = 0$ (C) $f'(x) = x^2 + 1$ (D) $f'(1) = 2$

MDY146

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Integer Type Question

6. Let $f(x) = \min\{|x-1|, |x+1|, 1\}$. Find the number of points where it is not differentiable.

MDY147

7. Consider function $f(x) = \min\{\tan x, \cot x\}$ in $(0, 2\pi)$. Number of points where f is either fails to be derivable is $'m'$ and number of points where it is discontinuous is $'n'$. Then the value of $m - n$ is

MDY148

8. Let $f(x) = \max\{2\sin x, 1-\cos x\}\ \forall\ x \in (0, \pi)$. If $f(x)$ not differentiable at $x = \pi - \cos^{-1}\frac{k}{5}$ then the value of k is

MDY149

9. $f(x) = \begin{cases} |1-4x^2| & , \quad 0 \le x < 1 \\ [x^2 - 2x] & , \quad 1 \le x < 2 \end{cases}$ then number of points where $f(x)$ is non-derivable in $x \in (0, 2)$ is equal to

MDY150

10. $f(x) = |[x]\, x|$ then number of points where $f(x)$ is non-derivable in $x \in (-1, 2]$ is equal to (where [.] denotes greatest integer function)

MDY151

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	A	B,C	B,C	A,B,C,D	5	4	3	2	3

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